

The Degree of Saturation of a Solution

PURPOSE

- To prepare aqueous solutions of sodium acetate with different degrees of saturation.
- To study the properties of these solutions.

INTRODUCTION

A solute can dissolve in a given amount of solvent to give three different kinds of solution. The resulting solution can be unsaturated (contain less solute than the solvent is capable of holding), saturated (contain the maximum amount of solute that the solvent is capable of holding), or supersaturated (contain more than the amount of solute that the solvent is normally capable of holding under the given conditions).

In this experiment, you will prepare solutions of each type, identify them, and study some of their properties.

APPARATUS

150 mm test tubes (2)
rubber stopper
test tube holder

burner
250 mL beaker
glass stirring rod

MATERIALS

hydrated sodium acetate

PROCEDURE

- Place 5 mL of distilled water in a 150 mm test tube. Add a few milligrams (a few tiny crystals) of hydrated sodium acetate. Close the test tube with a rubber stopper and shake it vigorously for 30 s. Add a few more crystals and shake the test tube again for another 30 s.
- Now add 8.0 g of hydrated sodium acetate to the solution. Stopper the test tube and shake it vigorously for 1 min. Wait 1 min to allow settling of any undissolved matter. Use a glass stirring rod to push any solid that is above the liquid down into the liquid. Pour the clear upper liquid into a *dry* test tube. To the clear liquid in the second test tube add a few crystals of sodium

acetate, stopper, and shake vigorously for 1 min. Make very careful observations of the test tube.

- C. Allow the cold water tap to run so that at the end of part C you will have the coldest water possible. Pour the contents of the second test tube back into the first tube and add an extra 2.0 g of hydrated sodium acetate. Holding the test tube in a nearly horizontal position and *taking care not to point the open end at anyone*, heat the liquid *gently* (by moving the test tube in and out of the flame) to dissolve the solid present. Avoid "bumping" (sudden irregular bubbling) which might cause you to lose some of the contents of the test tube. Use a glass stirring rod to push as much solid at the top of the test tube down into the liquid. Continue the gentle heating until *all* of the solid has dissolved.
- D. Gently place the test tube in a beaker of cold water and allow it to remain undisturbed for 5 min. If crystals appear at any point, remove the test tube, reheat it gently to redissolve the crystals, and place the test tube in water for another 5 min. Gently remove the test tube and add one or two tiny crystals of sodium acetate.

EXAMINING THE OBSERVATIONS

1. What kind of solution did you have in part A? How do you know?
2. What kind of solution was the clear liquid that you added to the second test tube in part B? How do you know?
3. What kind of solution did you have in part D before the crystals of sodium acetate were added? How do you know?
4. What kind of solution did you have in part D after the crystals were added?
5. What happens when a small particle of solute is added to an unsaturated solution? Explain.
6. What happens when a small particle of solute is added to a saturated solution? Explain.
7. What happens when a small particle of solute is added to a supersaturated solution? Explain.

FOR FURTHER THOUGHT

8. Why did your second test tube in part B have to be dry?
9. Devise a test to determine whether a solution is unsaturated, saturated, or supersaturated.
10. Would it be possible to form a supersaturated solution of a compound whose solubility decreases with increasing temperature? Explain.
11. Does a solution equilibrium exist in a supersaturated solution? Explain.